Büchli: checking campaign claims against the official voting booklet

Multilingual document-grounded NLI with Apertus v1.5 — Hack Apertus 2026, Track 2A (OST)

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Code, demo, run files and all predictions in the repository; image `ghcr.io/kaiser-data/hack-apertus-booklet-nli:latest`

Abstract

Swiss voters meet campaign claims in many languages, while the authoritative source is the official voting booklet. We present *Büchli*, a system that decides whether a booklet (German, French or Italian PDF) *supports*, *contradicts* or *does not settle* a claim in any of the three languages, and quotes booklet sentences with their pages. A small CPU embedder selects 16 passages and ranks quotations; Apertus v1.5 is the only model that decides and cites. On 976 held-out test claims, Apertus-v1.5-8B reaches macro-F1 **0.891** (95% CI 0.871–0.909) with ∼3,400 prompt tokens per claim; on a paired 60-claim subsample it scores 0.861 against 0.735 for reading the whole booklet at 31,400 tokens. Under the organisers' scorer on 150 dev claims per task, the same pipeline with Apertus-v1.5-70B reaches **0.905–0.925** with booklets and **0.979–0.986** with given passages (two development runs). The repository includes a demo, hosted publicly during the hackathon, and an explorer of all 976 test answers. Of three evidence configurations, sentence quotes match the annotated passages best (0.758, against 0.535 for whole pages).

1 Introduction

Swiss voters decide several times a year on complex legal, social and financial proposals, and during each campaign they meet claims that may or may not match the official booklet, often in another language than the one they read. An informed vote depends on checking such a claim against the source, and a language model that misreads it, for example a legal sentence with a qualification or a negation, is a real risk for democratic decisions. We call the system *Büchli*, the Swiss name for the voting booklet (*Abstimmungsbüchli*) that every household receives. A typical case: a French claim says a tax applies from a turnover of 500 million euros, while the German booklet says 750 million (p. 21); Büchli answers *contradicts* and quotes that sentence (more cases in Appendix A). We ask three research questions:

RQ1 Does Apertus understand a booklet across German, French and Italian, without translation?
RQ2 How much of the booklet must it read, and does it find the passage that settles the claim?
RQ3 At what granularity (pages, passages, sentences) should evidence be quoted?

Contributions. (1) A retrieval-then-verify pipeline in which Apertus is the only decision model, packaged for the organisers' evaluation contract. (2) A leakage-free split of the OST benchmark by vote date, with a single held-out test run. (3) Controlled comparisons of context size, evidence granularity, model size (8B vs 70B) and embedder cost, scored with the organisers' own scorer. (4) A demo and an explorer of all 976 test answers with their cited pages (Fig. 1; code in `modal/demo.py`, explorer `docs/results.html`). (5) All predictions, run files and figures published for re-analysis.

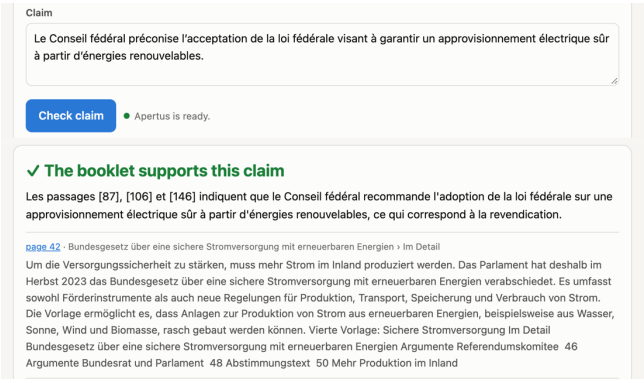


Figure 1: **The demo** (`modal/demo.py`; hosted at booklet-nli.pages.dev/demo during the hackathon): a French claim against the German booklet of 9 June 2024. Apertus answers that the booklet *supports* it and cites page 42, which can be checked in the official PDF. A user picks a booklet, types a claim in German, French or Italian and gets a verdict in about 5 s; the explorer (`docs/results.html`, works offline) lists all 976 recorded test answers.

2 Related work

NLI classifies a hypothesis as entailed, contradicted or neutral given a premise [1], across languages in XNLI [2]; FEVER [3] verifies claims against retrieved evidence sentences. Our premise is a whole booklet (16–144 pages) in another language than the claim. We follow retrieval-augmented generation [4] with a multilingual sentence embedder [5]; layout-aware parsing (Docling [6]) and visual page retrieval (ColPali [7]) are alternatives we test in Appendix B.

3 Task and data

Task. Document + claim → entailment / neutral / contradiction, judged against the booklet only, not real-world truth. The official evaluation runs our Docker image on a private set with two tasks: *A*, booklet PDF + vote name + claim; *B*, a given reference passage + claim [8, 9]. All nine claim/booklet language pairs occur. **Data.** The OST benchmark [10] has 1,488 pairs with approximately balanced labels, covering all nine language pairs (147–196 each); booklets come from the Federal Chancellery [11]. It ships without a split, so we split *by vote date*, keeping the three language versions of a booklet on the same side: dev 6 dates / 512 pairs (all design choices), test 14 dates / 976 pairs (run once).

4 Method

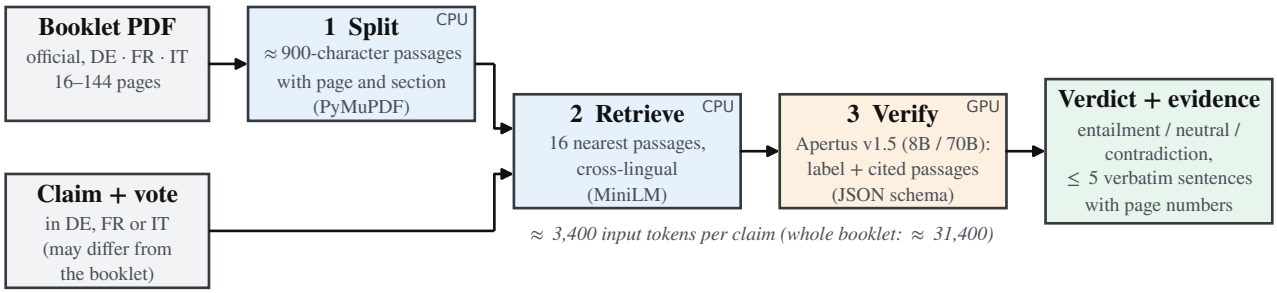


Figure 2: **The Büchli pipeline.** Booklet and claim may be in different languages; only the 16 passages closest to the claim reach Apertus, which returns the label and the passages it relied on; verbatim sentences with their pages are quoted as evidence. Editable source: `paper/pipeline.excalidraw` (open at excalidraw.com).

Steps (Fig. 2). *Split:* PyMuPDF [12] extracts each page, rejoins words broken across lines and merges text blocks into passages of about 900 characters (195 of 7,610 longer) that never cross a page and keep the PDF outline path (vote → part → subsection). *Search:* `paraphrase-multilingual-MiniLM-L12-v2` [5, 13] (fastembed [14]) embeds claim and passages in one cross-lingual space; the 16 nearest go to Apertus in document order. *Decide:* one call, temperature 0, JSON schema {`reason`, `evidence`, `label`}; the prompt defines the labels, names the vote and treats a statement attributed to the wrong actor as a contradiction. *Quote:* up to five verbatim sentences with pages, three from the cited passages, then two from the rest of the context. The Docker image follows the evaluation contract (CPU only, no runtime downloads) and uses Apertus-v1.5-70B [15] when served, else the 8B [16].

5 Experimental setup

Metrics. Macro-F1 over the three labels (primary); evidence Hit@5 with the organisers' matcher (a quote counts if it fuzzily lies inside the gold passage or contains it); input tokens per case; non-LLM processing time. 95% intervals from 2,000 bootstrap resamples of claims. **Official scoring** [9]. Valid only if macro-F1 ≥ 0.60 (A) and ≥ 0.75 (B; slide 16, while the starter's `evaluate.py` prints 0.70). Task A $= 0.5\,\text{F1} + 0.2\,\text{Hit@5} + 0.15\,\text{time} + 0.15\,\text{tokens}$; task B $= 0.7\,\text{F1} + 0.15\,\text{time} + 0.15\,\text{tokens}$; final $= 0.6A + 0.4B$; time is wall-clock minus time with an LLM request in flight, and both efficiency terms are relative to the best team. **Baselines.** Whole booklet in the prompt; $k = 8$ passages; two alternative neural embedders and BM25 fusion.

Component	Tool / model	Where it runs
PDF parsing	PyMuPDF 1.28	CPU, in the image
Retrieval	paraphrase-multilingual-MiniLM-L12-v2 (ONNX, fastembed 0.9)	CPU, weights baked into the image
Decision model	Apertus-v1.5-8B; Apertus-v1.5-70B (submitted default)	vLLM [17]: 1×L40S (8B), 2×H200 (70B), Modal
Scoring	organisers' `evaluate.py` [8], commit 5e69577	unmodified
Embedders compared	multilingual-e5-large [18], Qwen3-Embedding-0.6B [19], BM25 + RRF [20, 21]	4 vCPU, retrieval only

Table 1: Methods and tools. Versions are pinned in `uv.lock`; the vLLM image is pinned by digest.

6 Results

6.1 Main results

Split	Model	Task A F1	Task B F1	Evid. (A)	In tokens A / B	Out A
Test, 976 (own scorer)	8B	0.891 (0.869–0.913)‡	–	0.716†	3,424 / –	86
Dev, 150/task (official)	8B	0.857	0.930	0.768	3,349 / 2,286	86
Dev, 150/task (official)	70B, run 1	**0.925**	0.979	**0.808**	3,349 / 2,286	92
Dev, 150/task (official)	70B, run 2	0.905	**0.986**	**0.808**	3,349 / 2,286	92

Table 2: **Main results.** "Official": the organisers' scorer on cases in their format, run in our Docker image. †Test evidence uses our own n-gram overlap measure. ‡Bootstrap over the 14 vote dates (claim-level: 0.871–0.909). A near-repeat of the 8B dev run (larger output cap) moved F1 by 0.027. Files: `runs/test-rag-k16-n976/`, `runs/contract-dev-n150/`.

RQ1, languages. On test, cross-lingual pairs score 0.899 (n = 641) and same-language pairs 0.875 (n = 335): no *average* penalty. The average hides a weakness (Fig. 3): on contradiction claims that mention an actor, the 8B errs in 14 of 107 cross-lingual cases against 2 of 41 same-language ones (small numbers). Claims say "the committee", "le comité" or "il comitato", while the booklet names a specific body in another language ("comitati referendari", "Initiativkomitee"); in 11 of the 14 the right passage was in context and the model answered *neutral*. Reading these cases suggests that mapping actors across languages, rather than translation as such, is the open problem. A prompt glossary of actor names across the three languages did not fix it (actor claims 82/90 before and after with the 8B; 70B task B 0.958 against 0.979–0.986, at +137 tokens), so it is not used.

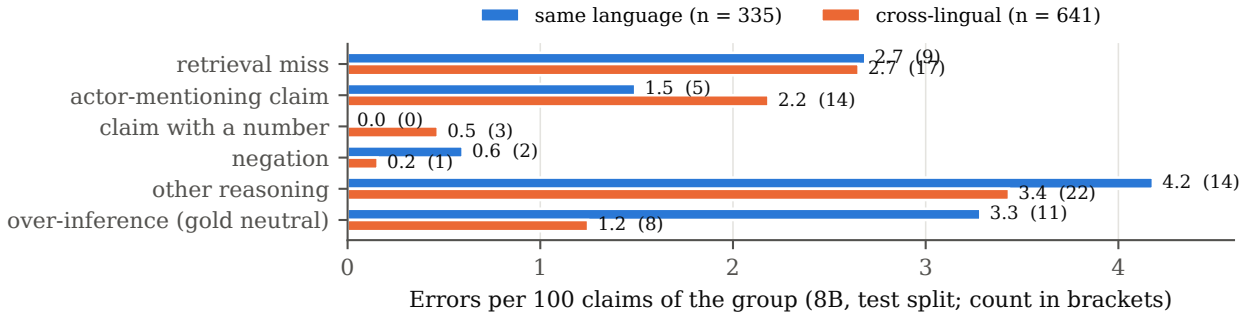


Figure 3: **Where the 8B fails**: all 106 test errors, sorted by keyword rules applied in order (`paper/figures.py`), per 100 claims of each language group. The categories describe the claim (e.g. it mentions an actor or a number), not a verified cause. Retrieval misses hit both groups equally; errors on actor-mentioning and number claims are more frequent across languages; over-inference (supporting or contradicting a claim the booklet does not settle) is more frequent within one language.

RQ2, context. Sixteen retrieved passages beat the whole booklet at a ninth of the tokens (Fig. 4). Halving the context ($k = 8$) saves 45% of tokens but costs F1 (8B dev 0.850 vs 0.890) and evidence (70B 0.677 vs 0.828), so we keep $k = 16$. Given the annotated passage instead (task B), the 8B scores 0.930 against 0.857: of its

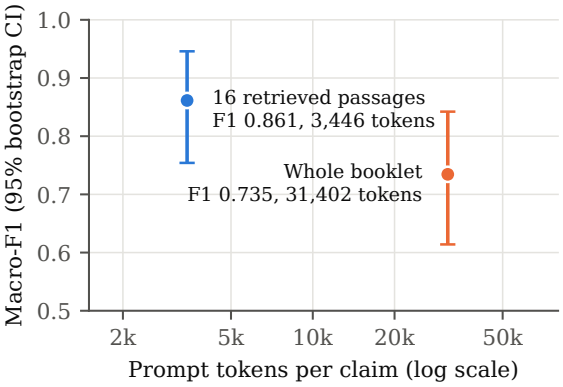


Figure 4: **Higher measured F1 at a ninth of the input tokens** (RQ2). Same 60 held-out test claims, 8B. The whole booklet costs 9× the tokens and six of its answers never finish (the model lists passage numbers until the output cap); excluding those, retrieval still leads by +0.053 (CI −0.056 to +0.168). Bars: 95% bootstrap intervals. Source: `paper/figures.py`.

21 task-A errors, 14 disappear, 7 remain and 3 new ones appear, so errors persist even with the reference text; separating retrieval, context interference and reasoning needs case-level analysis. Contradiction read as neutral is the main confusion (31 of 106 test errors); retrieval misses explain 26 of the 106.

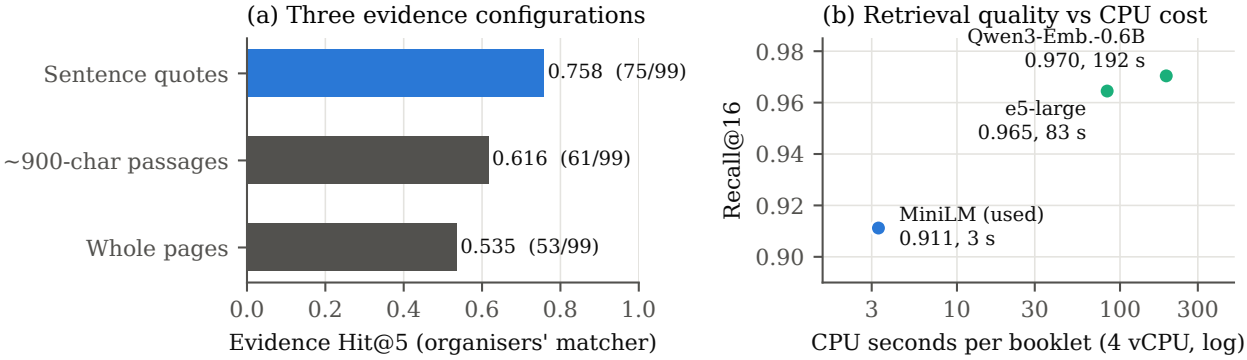


Figure 5: **(a) The scorer rewards short quotes (RQ3); (b) retrieval quality versus CPU cost.** (a) Three evidence configurations (8B, 99 non-neutral dev claims, official matcher); pages and passages share one run, sentences come from a second run. A gold passage often includes a heading that a page does not reproduce contiguously. (b) Recall of the annotated passage in the top 16 (338 dev claims) against CPU seconds per booklet on 4 vCPU. Sources: `runs/contract-dev-n150/evidence-variants/`, `runs/embed-bench-dev.json`.

RQ3, evidence. Three evidence configurations score 0.535 (whole pages), 0.616 (passages) and 0.758 (sentences) under the official matcher (Fig. 5a). The comparison is not fully controlled: the sentence configuration comes from a separate inference run (two task-A labels differ) and also adds sentences from uncited context. **Model size.** With identical prompts, retrieval and input tokens (output 92 against 86), the 70B gains 5–7 points on the booklet task and 5–6 on given passages, in two dev runs; held-out confirmation remains (Table 2). **Embedder cost.** Larger embedders retrieve better (recall@16 0.965 and 0.970 against 0.911) but need 83 and 192 s of CPU per booklet against 3 s; BM25 fusion did not help at $k = 16$ (Fig. 5b). In a task-A-only container the submitted image spends 0.22 s of non-LLM time per case. **Inference cost.** One Apertus call per claim; end-to-end median 3.9 s (8B) and 4.9 s (70B) per booklet claim. At Modal list prices and our measured throughput (warm server, 6 parallel requests) the GPU costs about $0.46 per 1,000 claims for the 8B (one L40S) and $3.75–6.48 for the 70B (two H200).

7 Discussion

What we learned. On this benchmark Apertus works across the three languages without a translation step on average, but cross-lingual actor attribution remains a weak spot (RQ1). The bigger model is better on our dev runs, as expected. Errors remain even when the 8B has the annotated text in hand, which we think fine-tuning can address. The scorer, not intuition, decides what good evidence is: our first instinct, and that of another AI we asked, was to cite whole pages; measuring with the organisers' matcher showed that single sentences work best. **Hackathon scoring versus real use.** The scorer charges the LLM only for tokens (equal for 8B and 70B), not for waiting time, but charges CPU time; hence the 70B and the small embedder. A real service

would differ: users wait for each answer and every 70B query costs GPU time (the 8B answers in a median 4.5 s at 0.891 on test), while booklets could be embedded once ahead of time, which removes most of the cost of stronger embedders. Retrieving instead of reading everything holds in both settings; the evidence result is specific to the reference-matching metric. **Limitations.** A near-repeat of the 8B dev run moved macro-F1 by 0.027 (batched GPU inference is not bit-exact); decisions rest on 6 dev vote dates and the 70B result on dev runs only. Hit@5 is a proxy, and two of five quotes come from context the model did not cite (the demo shows only cited passages). Retrieval can pick the wrong vote in a multi-vote booklet, and PyMuPDF reads tables as lines.

8 Ethics, use of AI and reproducibility

Ethics. A verdict states what the booklet says, not whether a claim is true or how to vote; it is a model prediction to check against the cited page, not political advice. The benchmark contains no personal data; the live demo processes visitor IP addresses only to enforce daily rate limits and does not store claims. **Use of AI.** Apertus-v1.5 predicts labels and cites passages; MiniLM ranks passages and quotations. Development used Claude Code (Anthropic, Claude Opus 5.5), which implemented the author's ideas in code, ran the experiments on Modal and drafted documentation under the author's direction; read-only reviews came from Codex CLI (OpenAI, five), Grok (xAI, two) and one further AI review. Every finding was checked against the code; scores, token counts and figures are recomputed from `runs/`. **Reproducibility.** Code, run files and all predictions are in the repository (Apache-2.0); dataset revision `fc2b276`; temperature 0, seed 0. Every experiment, including rejected ones (prompt changes, parser v4, vote routing, a $k = 8 \rightarrow 16$ cascade), is documented in the archival `technical_report.md`.

9 Conclusion and future work

Apertus checks claims against the official booklet across German, French and Italian, at a ninth of the whole-booklet tokens, and shows the page that settles them. Next: test many more claims and all language combinations; fine-tune by distilling the 70B into the 8B (teacher–student) or with a LoRA adapter, on booklets outside the benchmark and split by vote date; add an actor-aware verification step (a prompt glossary alone did not help), tested on minimal pairs that change only the speaker or a number; and move retrieval to page images: in a first test, ColQwen2 page retrieval finds the gold page in the top 5 for 93.5% of dev claims against 71.3% for our text pipeline (Appendix B). Beyond the benchmark, the same approach can serve the many voters who never read the booklet, answering "what does this proposal actually say about X?" with the cited page, ground opinion surveys and political barometers in the official text, and extend to cantonal and communal votes and to other countries, for example Germany.

A voter can only decide well if a claim can be traced to the page that settles it; Apertus does that across German, French and Italian.

References

[1] Samuel R. Bowman, Gabor Angeli, Christopher Potts, and Christopher D. Manning. A large annotated corpus for learning natural language inference. In *Proceedings of EMNLP*, 2015.

[2] Alexis Conneau, Ruty Rinott, Guillaume Lample, Adina Williams, Samuel R. Bowman, Holger Schwenk, and Veselin Stoyanov. XNLI: Evaluating cross-lingual sentence representations. In *Proceedings of EMNLP*, 2018.

[3] James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. FEVER: a large-scale dataset for fact extraction and VERification. In *Proceedings of NAACL-HLT*, 2018.

[4] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in NeurIPS*, 2020.

[5] Nils Reimers and Iryna Gurevych. Making monolingual sentence embeddings multilingual using knowledge distillation. In *Proceedings of EMNLP*, 2020. paraphrase-multilingual-MiniLM-L12-v2.

[6] Christoph Auer et al. Docling technical report. *arXiv:2408.09869*, 2024.

[7] Manuel Faysse, Hugues Sibille, Tony Wu, Bilel Omrani, Gautier Viaud, Céline Hudelot, and Pierre Colombo. ColPali: Efficient document retrieval with vision language models. *arXiv:2407.01449*, 2024.

[8] OST AI Applications & Deployment Lab. hackapertus-starter: CLI contract, `prepare_cases.py` and `evaluate.py`. `https://gitlab.com/ifsoftware/hackapertus-starter`, 2026. Commit 5e69577; `evaluate.py` used unmodified for the official-format dev scores.

[9] Mitra Purandare and Abinas Kuganathan. HackApertus Track 2A: Q&A – evaluation, starter repo and demo app. Slides, 8 October 2026, Hack Apertus Getting Started Guide (`https://hackapertus.notion.site/track-2a-ost`), 2026. Slides 16–18: validity thresholds, scoring weights, processing-time definition, evidence matching.

[10] OST Eastern Switzerland University of Applied Sciences. MNLI over Swiss Voting Booklets, v1.1. `https://huggingface.co/datasets/OSTswiss/MNLIoverSwissVotingBooklets`, 2026. MIT licence; revision fc2b276, SHA-256 checked on download.

[11] Swiss Federal Chancellery. Erläuterungen des Bundesrates / Explications du Conseil fédéral / Spiegazioni del Consiglio federale. `https://www.bk.admin.ch/de/sammlung-der-abstimmungsbuechlein-seit-1978`.

[12] Artifex Software. PyMuPDF. `https://github.com/pymupdf/PyMuPDF`, 2026. Version 1.28; AGPL-3.0 or commercial licence.

[13] Sentence-Transformers. paraphrase-multilingual-MiniLM-L12-v2. `https://huggingface.co/sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2`. Model card; 128-token input limit.

[14] Qdrant. FastEmbed: lightweight embedding generation. `https://github.com/qdrant/fastembed`, 2026. Version 0.9, ONNX runtime on CPU.

[15] Swiss AI Initiative. Apertus-v1.5-70B. `https://huggingface.co/swiss-ai/Apertus-v1.5-70B`, 2026. Model card; snapshot 94888dd, official bf16 weights.

[16] Swiss AI Initiative. Apertus-v1.5-8B. `https://huggingface.co/swiss-ai/Apertus-v1.5-8B`, 2026. Model card; official bf16 weights served with swiss-ai's vLLM build.

[17] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of SOSP*, 2023.

[18] Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. Multilingual E5 text embeddings: A technical report. *arXiv:2402.05672*, 2024.

[19] Yanzhao Zhang et al. Qwen3 Embedding: Advancing text embedding and reranking through foundation models. *arXiv:2506.05176*, 2025.

[20] Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389, 2009.

[21] Gordon V. Cormack, Charles L. A. Clarke, and Stefan Büttcher. Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. In *Proceedings of SIGIR*, 2009.

A Example cases

Claim	What the booklet says	Verdict
FR claim, DE booklet: "…s'applique aux entreprises dont le chiffre d'affaires annuel consolidé atteint **500 millions** d'euros."	p. 21: "…einen konsolidierten jährlichen Umsatz von **750 Millionen** Euro erreicht."	**contradicts** (8B): a changed number
FR claim, IT booklet: "**Le Conseil fédéral** préconise le rejet de la loi …, estimant que les médias ne sont pas indispensables à la démocratie directe."	p. 55: "La nostra democrazia diretta non può fare a meno dei media." (Federal Council and Parliament recommend yes)	**contradicts** (8B): the Federal Council recommends acceptance and says direct democracy needs media
FR claim, IT booklet: "…la présente loi prend effet **immédiatement** après l'expiration infructueuse du délai référendaire …"	p. 70: "La presente legge entra in vigore il **1º gennaio del sesto anno successivo** alla decorrenza infruttuosa del termine di referendum …"	8B: supports (wrong); **70B: contradicts** in both runs: a legal qualification

Table 3: **A changed number, a wrong actor and a legal qualification**, each across languages. Test claims #1159 and #1277; dev case `v1.1-row-1174-A`. Quotes verbatim, shortened with "…".

B Layout-aware parsing and visual page retrieval

Method (338 dev claims, 18 booklets)	Hit@1	Hit@3	Hit@5	Recall@16	Cost per booklet
PyMuPDF passages + MiniLM (submitted)	0.317	0.586	0.713	0.911	3 s CPU
Docling items + MiniLM	–	–	–	0.923	45 s CPU + 3 s
ColQwen2 over page images	**0.568**	**0.849**	**0.935**	–	5.3 s GPU (L40S)

Table 4: **Visual page retrieval finds the evidence page far more often.** Hit@k: a gold page in the top k (MiniLM scores a page by its best passage); Recall@16: the annotated passage in the top 16 chunks. ColQwen2 [7] (`vidore/colqwen2-v1.0`, 110 dpi), Docling [6] 2.135 without OCR; one run each. It needs a GPU at query time, so it is future work. Files: `runs/visual-bench-dev.json`, `runs/docling-bench-dev.json`.